

X sub-exponential $(2, \alpha)$

$$\mathbb{E}_X \left(e^{\lambda(X-\mu)} \right) \leq e^{\frac{v^2 \lambda^2}{2}} \quad \text{for } |\lambda| \leq \frac{1}{\alpha}$$

$$Z \sim \mathcal{N}(0, 1) \quad X = Z^2$$

Prove that X is sub-exponential $(2, \frac{4}{\alpha})$.

Sol. $\mu_X = \mathbb{E}(Z^2) = 1$

$$\mathbb{E} \left(\underbrace{e^{\lambda(Z^2-1)}}_f \right) = \int_{-\infty}^{+\infty} e^{\lambda(z^2-1)} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= \frac{e^{-\lambda}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{\lambda z^2 - z^2/2} dz \rightarrow e^{-\lambda} \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2} [1-2\lambda]} dz$$

$$= \frac{e^{-\lambda}}{\sqrt{2\pi}} \cdot \sqrt{2\pi} \frac{1}{\sqrt{1-2\lambda}} = \frac{e^{-\lambda}}{\sqrt{1-2\lambda}}$$

by def

$$\mathbb{E}(f(X)) = \mathbb{E}(Y) = \int_{-\infty}^{+\infty} y P_Y(y) dy$$

$$= \int_{-\infty}^{+\infty} f(x) P_X(x) dx$$

$$= \frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \quad \lambda < 1/2$$

Prove $\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{\frac{2\lambda^2}{2}}$ $|\lambda| \leq \frac{1}{4}$

$$\log \left(\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \right) \leq \log e^{2\lambda^2}$$

$$-\lambda - \frac{1}{2} \log(1-2\lambda) \leq 2\lambda^2$$

$$\underbrace{-\lambda - \frac{1}{2} \log(1-2\lambda) - 2\lambda^2}_{L(\lambda)} \leq 0$$

$$|\lambda| < 1/4$$

$$h(\lambda) \big|_{\lambda=0} = 0$$

$$h'(\lambda) = -1 - \frac{1}{2} \left(\frac{-2}{1-2\lambda} \right) - 4\lambda$$

$$= \frac{\cancel{1} + 2\lambda + \cancel{1}}{1-2\lambda} - 4\lambda$$

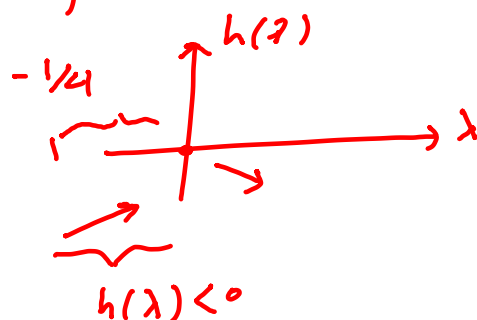
$$= 2\lambda \left(\frac{\cancel{1}}{1-2\lambda} - 2 \right)$$

$$= 2\lambda \left(\frac{1-2+4\lambda}{1-2\lambda} \right)$$

$$= \underbrace{2\lambda}_{?} \underbrace{(4\lambda-1)}_{?} \underbrace{\big/ (1-2\lambda)}_{>0}$$

$$|\lambda| \leq 1/4$$

$$\lambda \in [-1/4, 0) \rightarrow h'(\lambda) > 0$$



$$\lambda \in (0, 1/4] \rightarrow h'(\lambda) < 0 \Rightarrow h(\lambda) < 0$$

$$\Rightarrow \lambda \in [-1/4, 1/4] \Rightarrow h(\lambda) \leq 0$$

□

$$Z \sim N(0, 1) \quad Z^2 \text{ subexp}(2, 4)$$

prop. 2.9

if X subexp $(2, \alpha)$

$$\rightarrow P(|X - \mu| > t) \leq \begin{cases} 2e^{-t^2/2\alpha^2} & 0 \leq t \leq \frac{\alpha^2}{2} \\ 2e^{-t/2\alpha} & t > \frac{\alpha^2}{2} \end{cases}$$

$X_1, \dots, X_n$ sub-exp $(2, \alpha)$ indep. iid

$$P \left(\left| \underbrace{\frac{1}{n} \sum X_i}_{\text{empirical mean}} - \underbrace{\mu}_{\text{pop. mean}} \right| > t \right)$$

$$Y = \sum x_i$$

$$\mu_Y = \mathbb{E}(\sum x_i) = n\mu$$

$$\mathbb{P}(|Y - \mu_Y| > nt) \leq 2e^{-\frac{n^2 t^2}{2\sigma^2}}$$

Whether Y is sub-exp or not?

$$\begin{aligned} \mathbb{E}(e^{\lambda(Y - \mu_Y)}) &= \mathbb{E}(e^{\lambda(\sum x_i - n\mu)}) \\ &= \mathbb{E}\left(\underbrace{e^{\lambda(x_1 - \mu)}}_{\substack{\leq e^{\lambda^2 \sigma^2 / 2}}}\right) \cdot \underbrace{e^{\lambda(x_2 - \mu)}}_{\substack{\leq e^{\lambda^2 \sigma^2 / 2}}} \cdots \underbrace{e^{\lambda(x_n - \mu)}}_{\substack{\leq e^{\lambda^2 \sigma^2 / 2}}} \\ &= \prod_{i=1}^n \mathbb{E}(e^{\lambda(x_i - \mu)}) \leq \prod_{i=1}^n e^{\lambda^2 \sigma^2 / 2} \end{aligned}$$

$$\begin{aligned} &= e^{\lambda^2 / 2 \cdot \underbrace{(n\sigma^2)}_{(\sqrt{n}\sigma)^2}} \\ &= e^{\lambda^2 \sigma'^2 / 2} \end{aligned}$$

$$\begin{cases} \sigma' = \sqrt{n} \sigma \\ \alpha' = \alpha \end{cases}$$

Example 2.11

$$Z_k \sim \mathcal{N}(0, 1)$$

$$Z_k^2 \sim \text{Exp}(2, 4)$$

$$Y = \sum_{k=1}^n Z_k^2$$

$$\mathbb{E}\left(\frac{Y}{n}\right) = 1$$

$$\mathbb{P}\left(\left|\frac{1}{n} \sum Z_k^2 - 1\right| > t\right) \leq 2e^{-\frac{n t^2}{2 \cdot 4}}$$

$$\leq 2e^{-\frac{n t^2}{8}}$$

$$0 \leq t \leq \frac{4}{\sqrt{4}}$$

$$0 \leq t \leq 1$$

□

Johansen - Lindengrass.

$$F(u) := X u / \sqrt{m}$$

$$X_{m \times d}$$

$$X_{ij} \sim \mathcal{N}(0, 1) \text{ iid}$$

$$\forall i, j \quad 1 - \delta \leq \frac{\|F(u_i) - F(u_j)\|^2}{\|u_i - u_j\|^2} \leq 1 + \delta \quad \text{w.h.p.}$$

$$Y := \frac{\|Xu\|^2}{m\|u\|^2}$$

$$m \begin{bmatrix} \overbrace{x_1}^d \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} u \\ \vdots \\ u \end{bmatrix}_d$$

$$= \begin{bmatrix} \langle x_1, u \rangle \\ \langle x_2, u \rangle \\ \vdots \\ \langle x_m, u \rangle \end{bmatrix}$$

$$Y = \frac{\sum_{i=1}^m \langle x_i, u \rangle^2}{\|u\|^2} = \frac{1}{m} \sum_{i=1}^m \underbrace{\langle x_i, \frac{u}{\|u\|} \rangle^2}_{\omega_j}$$

claim: $\langle x_i, u/\|u\| \rangle \sim \mathcal{N}(0, 1)$ ✓

$$= \sum_j x_{ij} \cdot \frac{u_j}{\|u\|} \sim \mathcal{N}(0, \sum_j \omega_j^2)$$

claim 2. $\langle x_i, u/\|u\| \rangle$ & $\langle x_j, u/\|u\| \rangle$ indep.

$$\frac{1}{m} \sum_{i=1}^m \langle x_i, u/\|u\| \rangle^2 = \frac{1}{m} \sum_{i=1}^m z_i^2$$

$$\mathbb{E}(Y) = 1$$

$$\mathbb{P}(|Y - 1| > t) \leq 2e^{-\frac{mt^2}{8}} \quad t \in (0, 1)$$

$$\Rightarrow \mathbb{P}\left(\underbrace{\left| \frac{\|Xu\|^2}{m\|u\|^2} - 1 \right|}_{\text{not } \geq} > t\right) \leq \frac{2e^{-\frac{mt^2}{8}}}{1 - 2e^{-\frac{mt^2}{8}}} \quad t \in (0, 1)$$

w.h.p

$$-t \leq \frac{\|Xu\|^2}{m\|u\|^2} - 1 \leq t$$

$$1-t \leq \frac{\|Xu\|^2}{m\|u\|^2} \leq 1+t$$

$$1-t \leq \frac{\|F(\vec{u})\|^2}{\|u\|^2} \leq 1+t$$

$$1-2e^{-\frac{mt^2}{8}} \quad t \in (-1,1)$$

$$u \leftarrow u^i - u^j \quad B_{ij}$$

$$\mathbb{P} \left(1-t \leq \frac{\|F(u^i - u^j)\|^2}{\|u^i - u^j\|^2} \leq 1+t \right)$$

$$\forall i,j$$

$$2e^{-\frac{mt^2}{8}}$$

$$\mathbb{P}(\exists i,j \quad B_{ij} \notin (1-t, 1+t)) \leq \sum_{i,j} \mathbb{P}(B_{ij} \notin (1-t, 1+t))$$

$$\leq 2 \binom{N}{2} e^{-\frac{mt^2}{8}}$$

$$\Rightarrow \mathbb{P}(\forall i,j \quad B_{ij} \in (1-t, 1+t)) \geq \underbrace{1 - 2 \binom{N}{2} e^{-\frac{mt^2}{8}}}_{\text{w.h.p.}} \geq$$

$$\frac{N(N-1)e^{-\frac{mt^2}{8}}}{\log N(N-1) - \frac{mt^2}{8}} \leq \epsilon$$

$$\frac{m\delta^2}{8} \gtrsim 2 \log N - \log \epsilon$$

$$\rightarrow m \gtrsim \frac{16}{\delta^2} (\log N - \frac{1}{2} \log \epsilon)$$

$$\log N / \epsilon^{1/2}$$

□

$$\underline{1-\epsilon}$$